

Ching-Yu Hsu

Product Software Engineer

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Software engineer specializing in data engineering, AI systems, and ML infrastructure.

Education

University of California at Berkeley (UCB)

Aug. 2026 - May 2027

M.Eng. in Electrical Engineering and Computer Sciences (EECS)

California, U.S.

- Computer Systems concentration, Affinity Group Leader in Artificial Intelligence
- Relevant Coursework: Natural Language Processing (NLP), Advanced Topics in Computer Systems

National Taiwan University (NTU)

Sept. 2021 - Jan. 2026

B.S. in Electrical Engineering (EE)

Taipei, Taiwan

- Overall GPA 4.24/4.3, Overall Rank 5/268, Academic Excellence Awards, Presidential Award
- Relevant Coursework: Machine Learning, Deep Learning in Computer Vision, Data Structures, Algorithms

Technical Skills

Languages Python • C • C++ • Go • Java • JavaScript • TypeScript

AI / ML PyTorch • scikit-learn • Large Language Models (LLMs) • Vision Language Models (VLMs) • RAG • AI Agents

Backend FastAPI • Flask • Gin • Node.js

Infrastructure Docker • Kubernetes • Linux • Git • CI/CD • DevOps • Terraform • Kafka • n8n

Cloud & Data Amazon Web Services (AWS) • Google Cloud Platform (GCP) • SQL • MongoDB • Apache Airflow • dbt

Experience

NTUEE Programming Course, NTU

Aug. 2025 - July 2026

Organizer and Instructor

Taipei, Taiwan

- Designed and delivered lectures on LLM fundamentals and agent orchestration concepts for 500+ students
- Built an AI agent that searched for optimal defense against prompt injection attacks, enhancing safety by **30%**
- Developed an agentic system integrating Google Calendar and Gmail using Gemini, reducing manual effort by **80%**

CyberLink Corp.

July 2025 - Jan. 2026

Cloud Computing Research and Development Intern

Taipei, Taiwan

- Built a health-check agentic workflow using **n8n** for 300+ cloud servers, reducing diagnosis latency by **90%**
- Developed a production-level system supporting model orchestration across 20+ multi-modal generation pipelines
- Shipped production-ready features, driving the product from early prototyping through commercial release

Machine Learning Course, NTU

Jan. 2025 - Jan. 2026

Teaching Assistant

Taipei, Taiwan

- Designed AI programming assignments on LLMs and VLMs, and mentored students through office hours
- Developed and maintained an AI grading platform serving **1,100+** students, focusing on backend infrastructure, model evaluation, and GPU resource allocation

KKCompany Technologies

July 2024 - June 2025

Data Engineer Intern

Taipei, Taiwan

- Optimized an information retrieval system using **RAG** and LLMs, improving accuracy by 5% on an internal benchmark against state-of-the-art methods
- Built a data pipeline using distributed services including **Airflow** and **dbt**, enhancing system scalability by **95%**

Research and Project Experience

Vision and Learning Lab, NTU

July 2024 - Jan. 2026

Research Assistant

Taipei, Taiwan

- Designed an efficient multimodal AI system for visual question answering (VQA) through token pruning and motion-vector-based token selection, reducing GPU usage by over **90%** while maintaining competitive accuracy

Speech Processing and Machine Learning Lab, NTU

July 2023 - Jan. 2026

Research Assistant

Taipei, Taiwan

- Developed a co-evolutionary framework for LLMs to defend against prompt injection attacks, enhancing system reliability and increasing attack detection performance by 20%

CaRPG: in-car RPG game on smart windows

Mar. 2024 - June 2024

Full-Stack Engineer

Taipei, Taiwan

- Led a 5-member team to develop an AI-powered in-car RPG system, achieving 90% recognition accuracy and winning **1st Place of AUO Enterprise Award** at the 2024 MakeNTU Hackathon